

Wasiur R. KhudaBukhsh

List of publications

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In mathematics, authors are conventionally listed alphabetically. Links to all papers are available at www.wasiur.xyz/Publications.

PREPRINTS AND SUBMITTED PAPERS

1. Kushankur Dutta, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Coloured epidemic models: functional law of large numbers and propagation of chaos. 2026.
2. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, M. Gabriela M. Gomes, and Grzegorz A. Rempała. From individual-based stochastic epidemics to heterogeneous SIR equations. 2026. Submitted.
3. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Shrinkage priors for Bayesian substitute confounders. 2026.
4. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Fluctuation analysis of the standard QSSA for a stochastic multistage Michaelis–Menten model. 2026. Submitted.
5. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Self-intersection local times for Volterra Gaussian processes in stochastic flows with interaction. 2026. Submitted.
6. J. D. Harborne, **Wasiur R. KhudaBukhsh**, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Revision requested.
7. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Kernel-based estimators for functional causal effects. 2026.
8. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
9. Karim S. Elsayed, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.
10. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.

PEER-REVIEWED JOURNAL ARTICLES

11. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Statistical inference for a multiscale stochastic model of enzyme kinetics via propagation of chaos. *Stochastic Processes and their Applications*, 2027. Accepted for publication.
12. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
13. Yi Fu, Hye-Won Kang, **Wasiur R. KhudaBukhsh**, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi–Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.
14. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.

15. Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, **Wasiur R. KhudaBukhsh**, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
16. Yushuf Sharker, Zaynab Diallo, **Wasiur R. KhudaBukhsh**, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
17. **Wasiur R. KhudaBukhsh** and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
18. Matthew Wascher, Patrick Schnell, **Wasiur R. KhudaBukhsh**, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
19. István Z. Kiss, Luc Berthouze, and **Wasiur R. KhudaBukhsh**. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.
20. **Wasiur R. KhudaBukhsh**, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
21. **Wasiur R. KhudaBukhsh**, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in Ohio. *Journal of Theoretical Biology*, 561:111404, 2023.
22. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, and Grzegorz Rempała. Likelihood-free dynamical survival analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
23. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koepl. Hypergraphon mean-field games. *Chaos*, 2022.
24. **Wasiur R. KhudaBukhsh**, Casper Woroszylo, Grzegorz Rempała, and Heinz Koepl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.
25. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 2022.
26. Francesco Di Lauro*, **Wasiur R. KhudaBukhsh***, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-Markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Joint first authors.
27. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koepl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 2022.
28. Harley Vossler, Pierre Akilimali, Yuhan Pan, **Wasiur R. KhudaBukhsh**, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: study of 2018–2020 Ebola outbreak in Democratic Republic of the Congo. *Scientific Reports*, 12, 2022.
29. Ido Somekh*, **Wasiur R. KhudaBukhsh***, Elisabeth Dowling Root*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the population-level effect of COVID-19 mass vaccination campaign in Israel: a modeling study. *Open Forum Infectious Diseases*, 2022. *Joint first authors.
30. **Wasiur R. KhudaBukhsh**, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. Invited paper.
31. **Wasiur R. KhudaBukhsh**, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.

32. **Wasiur R. KhudaBukhsh**, Arnab Auddy, Yann Disser, and Heinz Koepl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 2019.
33. Hye-Won Kang*, **Wasiur R. KhudaBukhsh***, Heinz Koepl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *Joint first authors.
34. Saumya Yashmohini Sahai, Saket Gurukar, **Wasiur R. KhudaBukhsh**, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A machine learning model for nowcasting epidemic incidence. *Mathematical Biosciences*, 2021.
35. **Wasiur R. KhudaBukhsh**, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koepl. Generalized cost-based job scheduling in very large heterogeneous cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
36. Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, **Wasiur R. KhudaBukhsh**, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling outbreak data: analysis of a 2012 Ebola virus disease epidemic in DRC. *BIOMATH*, 8(2), 2019.
37. **Wasiur R. KhudaBukhsh**, Amr Rizk, Sounak Kar, and Heinz Koepl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 2019.
38. Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koepl, Boris Koldehofe, **Wasiur R. KhudaBukhsh**, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: a protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 2019.

BOOK CHAPTERS

39. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. *Handbook of Statistics*. Elsevier, 2024.
40. Grzegorz A. Rempała and **Wasiur R. KhudaBukhsh**. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

PEER-REVIEWED CONFERENCE PROCEEDINGS

41. Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the trade-off between fidelity and latency for the quantum link layer with few memories and entanglement purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QNCN 2024)*, 2024. Best Paper Award.
42. Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the fidelity distribution of link-level entanglements under purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
43. Riccardo Corradin, Luca Danese, Wasiur R. KhudaBukhsh, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
44. Wasiur R. KhudaBukhsh, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koepl. Collaborative uploading in heterogeneous networks: optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2018. <20% acceptance rate. Best-in-Session Presentation Award.
45. Wasiur R. KhudaBukhsh, Amr Rizk, Alexander Frömmgen, and Heinz Koepl. Optimizing stochastic scheduling in fork-join queueing models: bounds and applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2017. ~20% acceptance rate.
46. Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, 2017. Available: <http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf>.
47. Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 2017. ~26% acceptance rate. Best Paper Award finalist.
48. Wasiur R. KhudaBukhsh, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koepl. Analysing and leveraging client heterogeneity in swarming-based live streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 2016. ~26% acceptance rate.
49. Mahdi Mousavi, Hussein Al Shatri, Wasiur R. KhudaBukhsh, Heinz Koepl, and Anja Klein. Cross-layer QoE-based incentive mechanism for video streaming in multi-hop wireless networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 2017.

THESIS AND TECHNICAL NOTES

50. Wasiur R. KhudaBukhsh. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: <http://tuprints.ulb.tu-darmstadt.de/7588/>.
51. Mark Sinzger-D'Angelo, Heinz Koepl, and Wasiur R. KhudaBukhsh. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: <https://arxiv.org/abs/1808.00258>.